Lesson 1: Velocity and Time

Sample Worksheet - Marking Scheme

Physics - WPH11 (2018 specification)

Total: 20 marks. Original practice; not an official Pearson marking scheme.
Award equivalent valid physics. Each **[1]** is one distinct marking point. Allow consequential arithmetic when the subsequent method is correct, within the stated part maximum. Numerical answers require units.

Part A: MULTIPLE CHOICE QUESTIONS — 4 marks

Question	Answer	Notes	Marks
1	B	The velocity-time gradient is acceleration.	1
2	A	Areas below the axis count negatively for displacement.	1
3	C	Positive acceleration reduces the magnitude of this negative velocity.	1
4	D	Acceleration must be constant over the interval.	1

Part B: FILL IN THE BLANKS — 4 marks

Blank	Answer	Marks
1	vector	1
2	m s^{-2} / $\frac{\text{m}}{\text{s}^2}$	1
3	displacement	1
4	distance	1

Part C: STRUCTURED QUESTIONS — 12 marks

Question 5: From a graph to an equation — 7 marks

Part	Answer / marking points	Marks
(a)(i)	Uses $a = \frac{v-u}{t}$ **[1]**; rearranges $at = v - u$ to obtain $v = u + at$ **[1]**.	2
(a)(ii)	States displacement is the area under the velocity-time graph **[1]**; expresses this as rectangle plus triangle, $s = ut + \frac{1}{2}(v-u)t$ **[1]**; substitutes $v - u = at$ to obtain $s = ut + \frac{1}{2}at^2$ **[1]**. Accept trapezium area $s = \frac{1}{2}(u+v)t$ followed by substitution and expansion, with the same three ideas. A sketch is optional.	3
(b)(i)	Uses $s = 2.0(4.0) + \frac{1}{2}(1.5)(4.0)^2$ **[1]**; $s = 20$ m **[1]**.	2

Total for Question 5 = 7 marks

Part C: STRUCTURED QUESTIONS (continued)

Question 6: Direction and experimental uncertainty

5 marks

Part	Answer / marking points	Marks
(a)(i)	At a turn $v = 0$: $0 = -4.0 + 2.0t$, hence $t = 2.0$ s [**1**].	1
(a)(ii)	$s = ut + \frac{1}{2}at^2 = -4.0(6.0) + \frac{1}{2}(2.0)(6.0)^2 = +12$ m [**1**]. Accept 12 m rightwards.	1
(a)(iii)	The trolley travels 4.0 m left before the turn, then 16 m right. Total distance = $4.0 + 16 = 20$ m [**1**].	1
(b)(i)	Absolute uncertainty $\Delta a = \frac{0.88-0.72}{2} = 0.08$ m s^{-2} [**1**]; percentage uncertainty = $\left(\frac{0.08}{0.80}\right) \times 100\% = 10\%$ [**1**].	2

Total for Question 6 = 5 marks

Worked reasoning for the reversal

The velocity at 6.0 s is $v = -4.0 + 2.0(6.0) = +8.0$ m s^{-1}. The velocity-time line crosses zero at 2.0 s.
The area below the axis is a triangle of magnitude $\frac{1}{2} \times 2.0 \times 4.0 = 4.0$ m. It contributes -4.0 m to displacement.
The area above the axis is a triangle of magnitude $\frac{1}{2} \times 4.0 \times 8.0 = 16$ m. It contributes $+16$ m to displacement.
Therefore displacement $= -4.0 + 16 = +12$ m, while distance $= 4.0 + 16 = 20$ m. A zero velocity at one instant does not mean the acceleration is zero.

Interpreting the practical result

The gradient of the velocity-time line estimates acceleration. Using the specified half-range method, the result can be reported as $a = (0.80 \pm 0.08)$ m s^{-2}. The percentage uncertainty has no physical unit.
This acceptable-gradient method is a scaffolded extension. Repeating measurements may reduce the influence of random scatter; a systematic offset needs a calibration or zero check.

Part A	4 marks
Part B	4 marks
Question 5	7 marks
Question 6	5 marks
Total	**20 marks**

END OF MARKING SCHEME